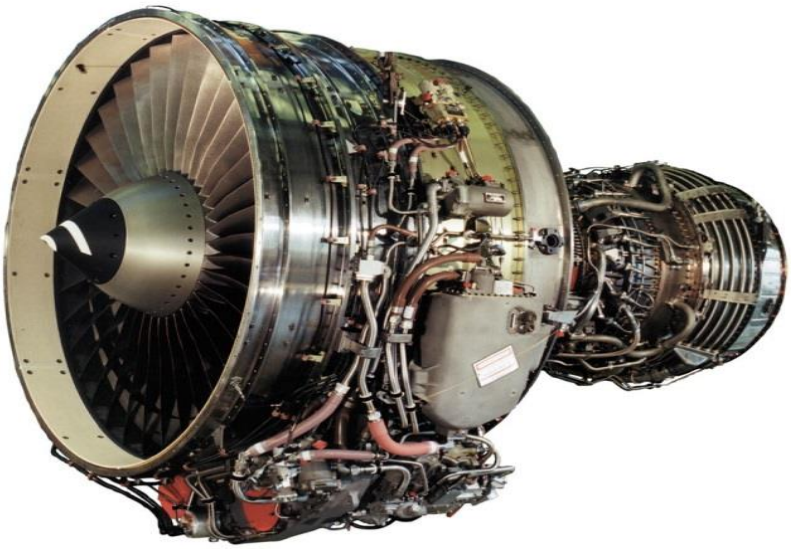
INTERNSHIP REPORT

Electrical and Control Systems of the CFM56 Aircraft Engine

Seyed Moein Razavi-Nabavi
Internship Supervisors: Dr. Sobhani and Dr. Ghorbani

Summer 2018



English translation prepared for academic portfolio use.
Not a certified translation.

Contents

- Abstract
- 1. Introduction
- 2. Main Components of a Turbofan and the CFM56
- 3. Engine Control Objectives and Operating Limits
- 4. CFM56-3 Control Architecture: MEC and PMC
- 5. Centralized and Distributed Control Architectures
- 6. Full Authority Digital Engine Control (FADEC)
- 7. Electronic Control Unit (ECU)
- 8. Engine Sensors: Overview
- 9. Pressure Sensors and Pressure Lines
- 10. Temperature Sensors
- 11. High- and Low-Pressure Shafts and Speed Sensors
- 12. Vibration Measurement and Accelerometers
- 13. Actuators and Primary Airflow Control
- 14. Conclusion
- References

Abstract

This report reviews the electrical and control systems installed on the CFM56 aircraft engine. The internship began with identifying the engine sensors and observing their locations on the engine, followed by research into their operating principles. The report then examines the Electronic Control Unit (ECU), the Full Authority Digital Engine Control (FADEC) system, and selected sensors and actuators. The sensor groups discussed include temperature, pressure, vibration, position, and shaft-speed sensors.

Keywords: CFM56; engine control; sensors; ECU; FADEC; actuators.

1. Introduction

Aircraft are commonly grouped into passenger and military aircraft. For a passenger aircraft, the engine is one of the most important systems. CFM International is a joint venture between General Electric (United States) and Safran (France); its engine family is known as CFM56.

The CFM56 is a turbofan engine. It produces thrust by accelerating gas and includes a fan, compressors, a combustion system, turbines, an exhaust system, and an integrated control system. In a turbojet, most of the intake air passes through the combustion chamber and exits as hot exhaust. In a turbofan, a substantial fraction of the air bypasses the combustion chamber and exits through the bypass duct. This architecture is well suited to the moderate flight speeds of many commercial aircraft.

Earlier engines relied more heavily on mechanical controls that allowed the pilot to influence fuel flow and engine power. Electronic control units made automatic control of many engine functions possible and supported better efficiency across changing operating conditions. The broader system responsible for these functions is called Full Authority Digital Engine Control (FADEC); the ECU is one of its principal components.

This report describes the ECU, FADEC, and selected electrical and control systems used on the CFM56 family, with particular reference to the CFM56-5C2 engine.

2. Main Components of a Turbofan and the CFM56

A turbofan engine can be described in terms of five principal sections:

Fan. Located at the inlet, the fan draws in a large volume of air. The report describes approximately 15% of the inlet flow as primary air and the remaining 85% as bypass air. Primary air enters the core, while bypass air contributes substantially to thrust and also helps cool the engine.

Compressors. The engine has low-pressure and high-pressure compressor sections behind the fan. They raise the pressure and temperature of the core airflow before combustion.

Combustion chamber. Compressed primary air mixes with fuel and burns. The report gives an approximate pressure increase of 30 times relative to the inlet and describes the gas temperature as reaching about 2,000 °F after combustion.

Turbines. High-pressure and low-pressure turbines extract energy from the hot gas. The high-pressure turbine drives the high-pressure compressor; the low-pressure turbine drives the fan and low-pressure compressor through their respective shafts.

Exhaust. The exhaust nozzle directs the gas flow rearward. The reaction to this flow generates forward thrust.

The CFM56-5C2 configuration discussed in the report includes one fan stage, four low-pressure compressor/booster stages, nine high-pressure compressor stages, a SAC combustor, one high-pressure turbine stage, and five low-pressure turbine stages.

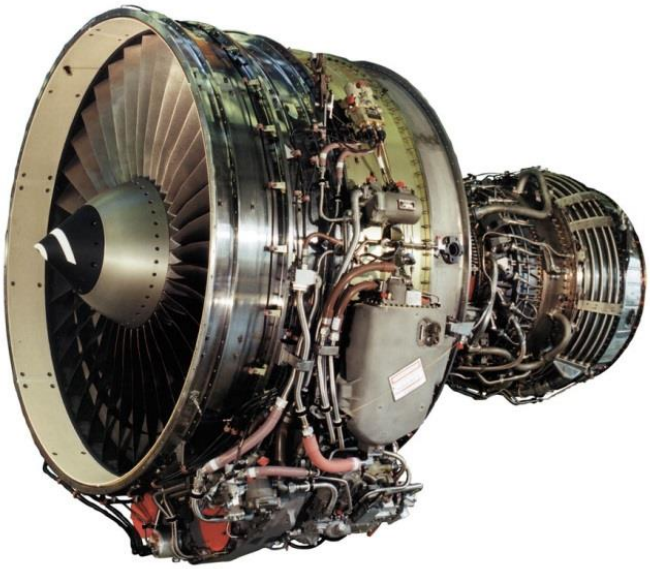


Figure 1. CFM56 engine, developed by the GE–Safran joint venture.

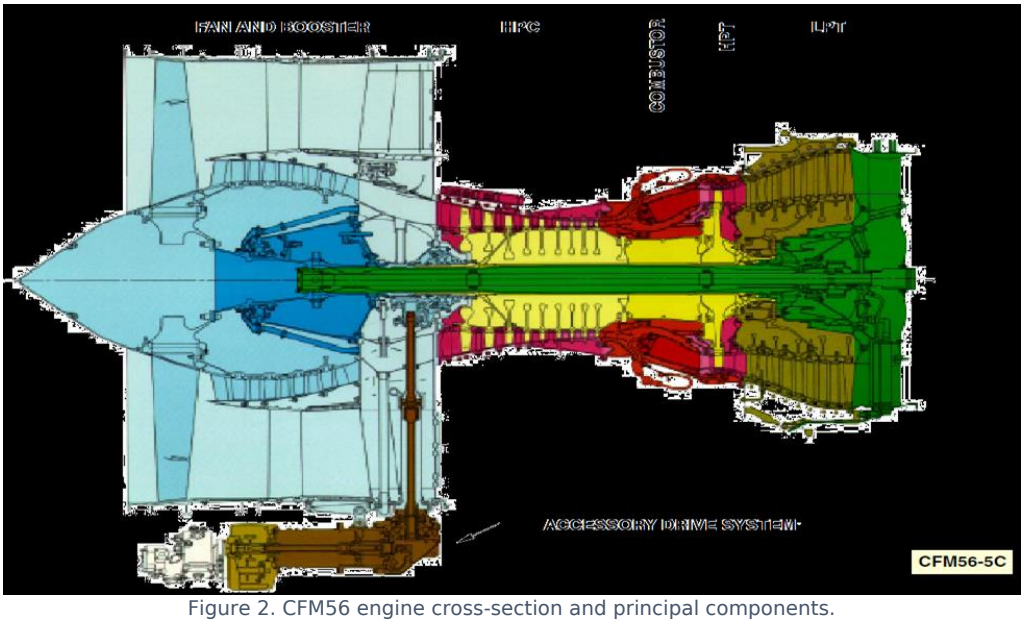


Figure 2. CFM56 engine cross-section and principal components.

3. Engine Control Objectives and Operating Limits

The report identifies three broad engine-control objectives: maintain a requested thrust for a given throttle-lever position; provide suitable performance as thrust demand changes; and ensure safe operation.

The first two objectives relate to power management. The third is addressed by protection logic, which prevents the engine from exceeding physical or operational limits. The report groups these limits as follows:

- Structural limits: maximum fan and N1/N2 shaft speeds, and maximum turbine-blade temperature.
- Safety limits: compressor and fan stall margins, and adequate fuel delivery to avoid incomplete combustion.
- Operational limits: a maximum turbine-inlet temperature intended to support turbine durability.

The engine control system must remain within a safe operating envelope. At low or high operating conditions, compressor stall or surge may occur, so fuel scheduling and airflow control must be coordinated with engine speed and pressure conditions.

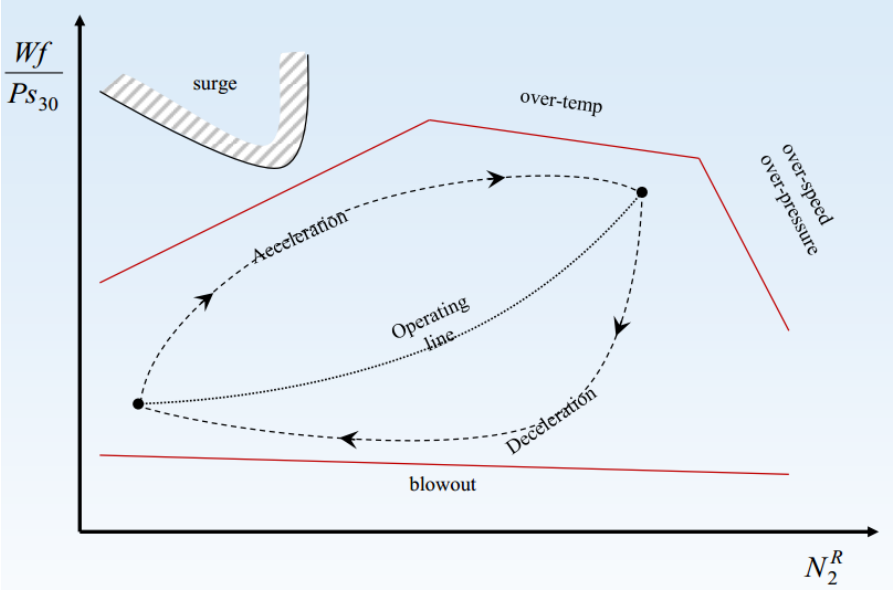


Figure 4. Example of engine operating limits and safe operating envelope.

4. CFM56-3 Control Architecture: MEC and PMC

The report describes the earlier CFM56-3 control arrangement as a combination of hydromechanical and electronic units. It is divided into Main Engine Control (MEC) and Power Management Control (PMC). The MEC is hydromechanical and operates using fuel pressure.

The primary MEC functions listed in the report are speed control, fuel control and limiting, and idle-speed operation. Additional functions include adjusting the Variable Bleed Valve (VBV), adjusting Variable Stator Vanes (VSV), and controlling high-pressure turbine clearance.

For a high-bypass engine, overall thrust is closely associated with fan speed N1; the report notes that the fan accounts for a large share of engine power. The PMC assists the pilot by automatically correcting the N1 target for altitude, pressure, and temperature. The original report includes schematic diagrams of the MEC/PMC arrangement and the CFM56-3 control system.

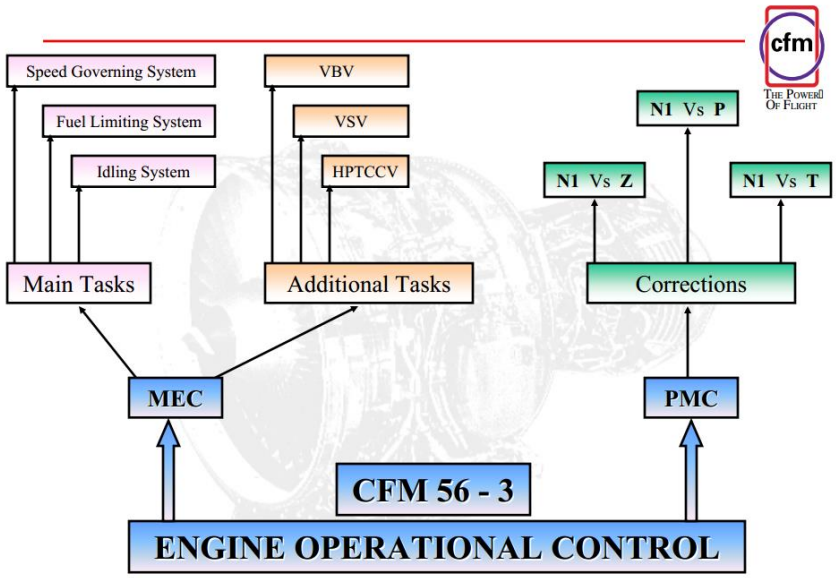


Figure 5. Schematic of the CFM56-3 engine control functions.

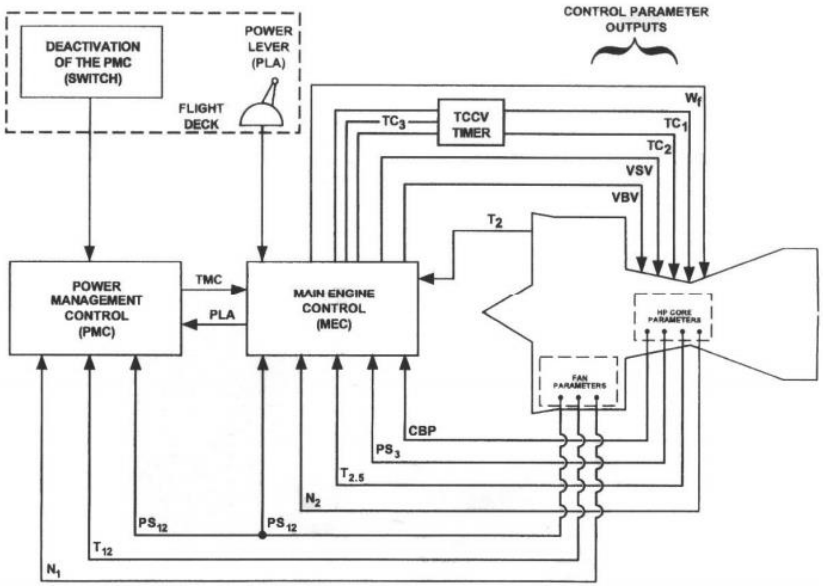


Figure 6. CFM56-3 engine control diagram.

5. Centralized and Distributed Control Architectures

Gas turbines offer high power-to-weight ratio, operational reliability, and relatively low emissions, and are used in aircraft, helicopters, and power-generation equipment. Further improvements in efficiency and lifecycle cost motivate the use of advanced active-control functions, including active combustion control and Active Clearance Control (ACC). Such functions can require intelligent sensors and actuators with substantial bandwidth.

A distributed control architecture connects sensors and actuators through a communication data bus rather than relying entirely on a centralized arrangement. The report discusses distributed control as a potential way to address issues such as communication delay and data loss, which can introduce uncertainty, reduce control quality, affect engine life, and increase lifecycle cost.

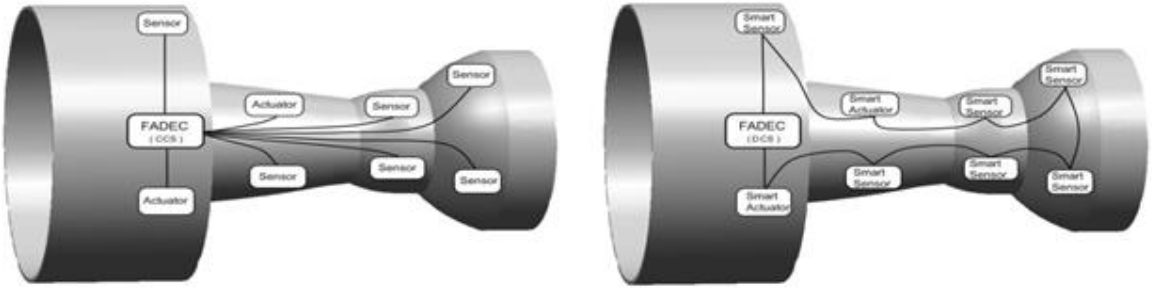


Figure 1.1: Centralized turbine engine control system

Figure 1.2: Distributed turbine engine control system

Figure 7. Centralized and distributed engine-control architectures.

6. Full Authority Digital Engine Control (FADEC)

FADEC is a digital engine-control system consisting of a digital computer—commonly called the ECU or Electronic Engine Controller (EEC)—and associated components. It controls major aspects of engine operation. Under a full-authority arrangement, the pilot cannot manually adjust the engine parameters that are assigned to the computer. The report notes that a total FADEC failure can lead to engine shutdown.

FADEC receives inputs such as air density, throttle-lever position, engine and ambient temperatures, and engine pressures, including total and static pressure. The ECU processes these inputs repeatedly and calculates commands for fuel flow, variable stator-vane position, bleed-valve position, and other controlled variables. The report states a processing rate of approximately 70 analyses per second and notes that FADEC also manages engine starting and shutdown.

The primary purpose of FADEC is to obtain efficient engine performance for the current flight conditions while enforcing programmed operating limits. It can also support engine-health monitoring and maintenance reporting.

Advantages identified in the report include automatic protection against operation outside limits; redundant channels; automatic thrust management; semi-automatic starting; better integration with aircraft systems; long-term health monitoring; monitoring/control of more parameters; fewer parameters requiring direct pilot monitoring; reduced weight; and improved fuel efficiency.

Disadvantages identified include limited manual intervention, possible loss of engine operation after a complete FADEC failure, greater system complexity than hydromechanical or manual systems, and the need for rigorous validation.

7. Electronic Control Unit (ECU)

The ECU is the operational core of FADEC. It receives sensor data, processes it, and issues commands to engine actuators to achieve the required operating condition. The report places the ECU on the fan case at approximately the 8:30 clock position and gives its weight as about 52 lb. Eleven electrical connectors and four pneumatic connectors connect the ECU to the aircraft and associated FADEC subsystems.

The ECU has two independent channels, A and B, each with its own computer and circuitry. One channel is active and controls valves and actuators; the other is on standby. They do not control the engine simultaneously. Cross-Channel Data Link (CCDL) communication provides corresponding information to both channels so each can perform its own processing.

The ECU selects the healthier channel according to fault-priority logic. The report lists critical faults such as processor, memory, and electrical failures, as well as faults affecting control of the Fuel Metering Valve (FMV), VSV, or VBV. The channels compare their information approximately every 30 ms, and the healthier channel is selected as active.

ECU interfaces are grouped into engine-sensor interfaces, aircraft interfaces, engine-control interfaces, and pneumatic engine-sensor interfaces. In addition to electrical inputs for temperature, speed, fuel flow, and thrust-reverser position, the ECU receives pressure inputs through probes and pneumatic lines. Pressure transducers inside the ECU convert these pressures into electrical signals. The report lists fan-discharge pressure Ps14, compressor-inlet pressure P2.5, compressor-discharge pressure Ps3, and low-pressure-turbine-inlet pressure P4.9, as well as fan-case static pressure P0.

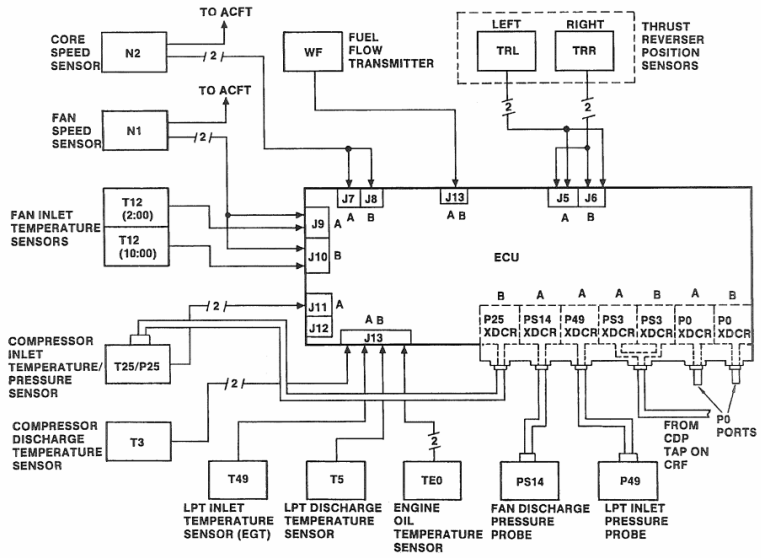


Figure 14. ECU and engine-sensor overview.

8. Engine Sensors: Overview

The engine sensors discussed in the report fall into temperature, pressure, position, vibration, and shaft-speed groups. Some measurements are displayed in the cockpit; others are used by the ECU to control actuators. The ECU contains A and B channels with similar functions, with one channel serving as a backup. The report states that the channel used is alternated between flights to avoid premature wear of one channel.

9. Pressure Sensors and Pressure Lines

According to the report, pressure at the engine measurement points is generally routed through tubing to the ECU rather than being converted directly at the measurement location. Transducers in the ECU calculate the pressure from the received signal. The report suggests that the tubing can reduce transient measurement error.

PS12. Located at the fan-frame inlet, PS12 measures local static pressure. The value is sent to the ECU, converted into an electrical signal, and used for engine control. Three probes are described at this location.

P0. The report states that this input is used by the ECU in calculating fuel-metering-valve position, VSV operation, and bearing/eleventh-stage cooling valves. The source text also describes P0 as measuring ambient temperature; this wording is retained here as reported, although P0 is elsewhere identified as a static-pressure parameter.

P2.5 / Ps14. These optional monitoring inputs measure bypass-air pressure in the fan frame. After conversion by the ECU, the values are sent to the cockpit for monitoring. The report assigns Ps14 processing to channel A and Ps2.5 processing to channel B.

Ps3. This input measures high-pressure-compressor discharge pressure. The ECU uses it in determining fuel-metering-valve and HPTACC operation; the report says the information is available to both channels and is also used for monitoring.

P4.9 / P49. This input measures pressure at the low-pressure-turbine outlet/inlet region as named in the source report; the original nomenclature is preserved.

Infinite-line pressure probe. The report cites NASA work on pressure lines used to measure high-frequency pressure fluctuations in demanding test conditions. A sufficiently long and properly designed line can reduce acoustic resonance effects that otherwise alter the frequency response of a pressure measurement system. Changes in cross-sectional area, particularly near the transducer, should be minimized. This arrangement can support high-frequency engine tests and small-diaphragm silicon transducers.

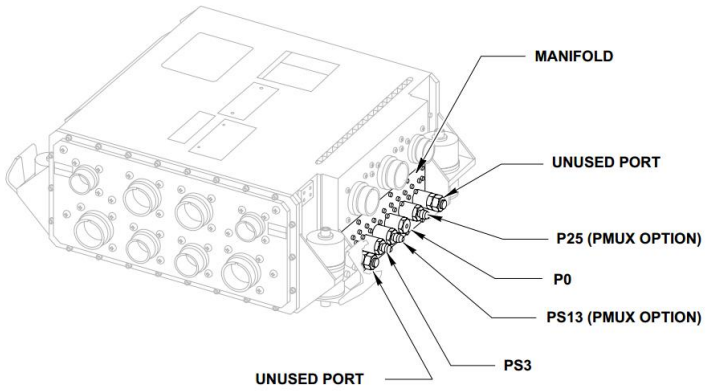


Figure 8. Pressure subsystem and pressure ports associated with FADEC.

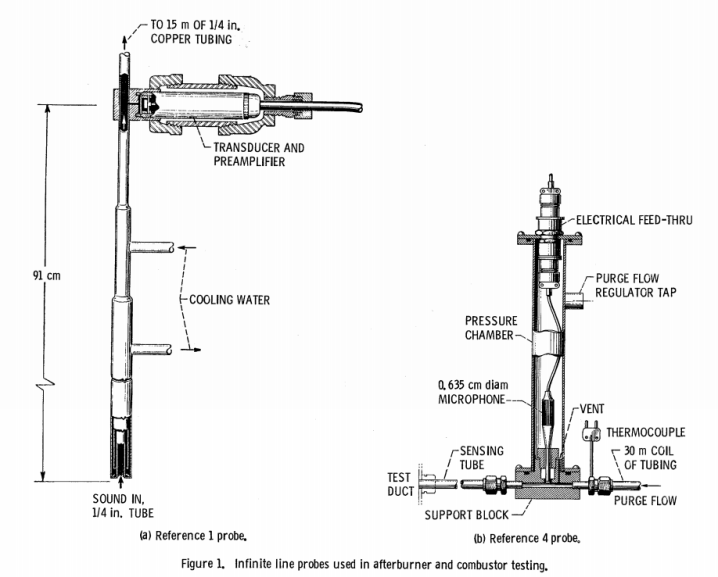


Figure 9. Infinite-line pressure probe concept.

10. Temperature Sensors

Temperature sensors are often installed in groups to improve measurement reliability. The report identifies Resistance Temperature Detectors (RTDs) and thermocouples as the two main types. Sensor names are based on their locations on the engine.

T12. An RTD at the fan inlet measures incoming-air temperature. The ECU supplies current and measures voltage across the sensor to determine resistance and infer temperature from the resistance–temperature relationship. The value is used for cockpit display and monitoring.

TP25. An RTD located on the fan frame, above the VBV and at the high-pressure-compressor inlet, is described as measuring temperature and pressure at the booster outlet/high-pressure-compressor inlet. The ECU uses this information for fuel control and to adjust VSV and VBV positions.

T3. A thermocouple located between the fuel-nozzle plate and the N flange measures high-pressure-compressor discharge-air temperature. A temperature difference between dissimilar metals generates a voltage that the ECU converts into a temperature estimate. The report says this measurement supports high-pressure-turbine clearance control.

T49.5. Thermocouples located in the exhaust-gas-temperature (EGT) region near the second low-pressure-turbine stage are described as a set of nine sensors. They provide temperature monitoring for the cockpit.

T5. A thermocouple at the low-pressure-turbine outlet measures local temperature for cockpit display and monitoring.

Toil. An RTD in the lubrication circuit provides oil-temperature information for ECU monitoring/control.

TCC. A thermocouple on the high-pressure-turbine case measures case temperature. FADEC uses the signal to regulate cooling flow through the HPTACC system.

11. High- and Low-Pressure Shafts and Speed Sensors

Shafts connect the turbines to their corresponding compressors. Some engines use multiple concentric shafts rotating at different speeds. In the CFM56 configuration described here, N1 denotes the low-pressure shaft and N2 the high-pressure shaft.

N1 speed sensor. The sensor uses a magnetic ring that rotates with the shaft. Tabs around the ring generate electrical pulses; one tab is larger and produces a stronger pulse that can be used as a reference when calculating shaft speed. The ring has 30 teeth. The sensor provides three analog electrical outputs through two connectors, generated by coils positioned behind a permanent magnet. One output is sent to the ECU and another supports aircraft vibration monitoring; the data can also be used to trim fan balance.

N2 speed sensor. Its operating principle is similar to the N1 sensor, but the ring has 71 teeth. The report places the sensor on the right-hand side of the accessory gearbox, inside the HMU. Outputs support the ECU and monitoring functions.

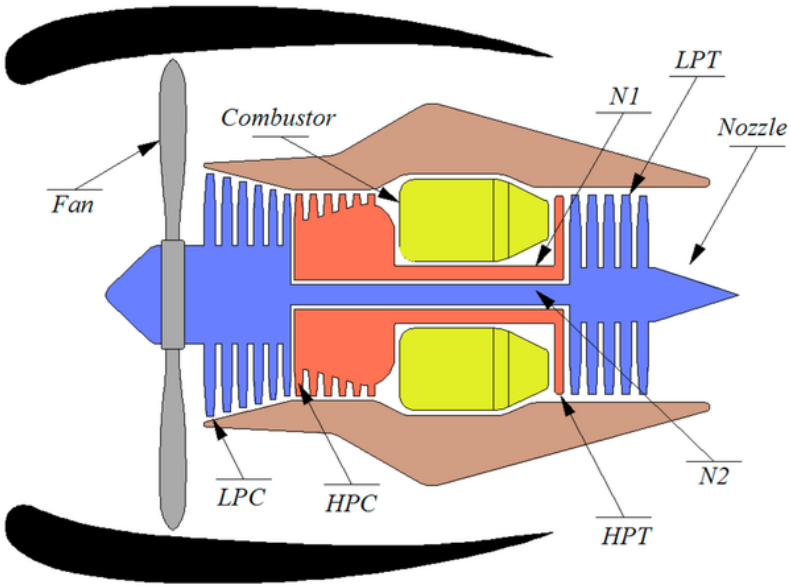


Figure 10. Turbofan layout showing low- and high-pressure shafts.

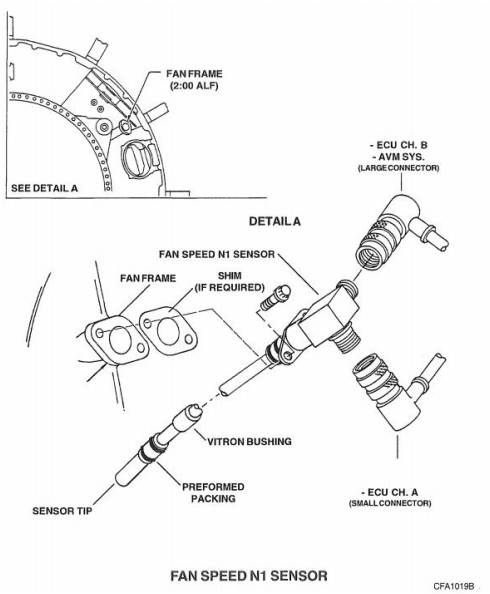


Figure 11. N1 speed-sensor location and components.

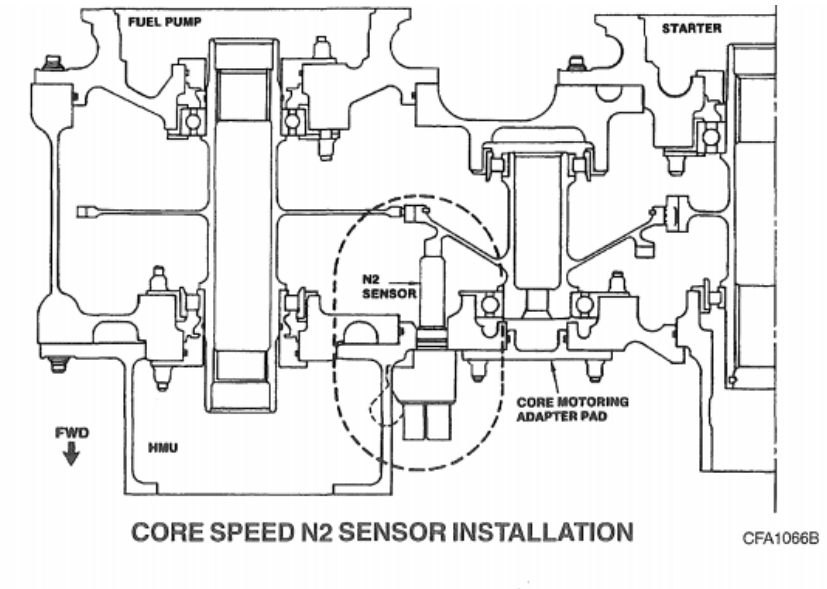


Figure 12. N2 core-speed sensor installation.

12. Vibration Measurement and Accelerometers

Vibration sensors provide measurements for cockpit monitoring and engine-condition assessment. The report introduces the dynamic model of a vibration instrument and distinguishes low-natural-frequency seismic instruments from high-natural-frequency accelerometers.

When the instrument natural frequency is high relative to the vibration frequency being measured, the instrument response can be related to acceleration. Different accelerometer designs are used for different frequency ranges. Moving-mass accelerometers are commonly used at lower frequencies; support springs may incorporate strain gauges connected in a bridge circuit. In force-balance designs, feedback holds the relative movement of the seismic mass close to zero, and the force required to do so indicates acceleration.

Piezoelectric accelerometers convert mechanical stress into electrical charge. A piezoelectric ceramic element is clamped against a seismic mass; vibration produces alternating mechanical forces on the element, generating charge proportional to acceleration. Sensitivity may be specified in mV/g or related acceleration units. Their high natural frequency makes them suitable for measuring high-frequency vibration in gears and rolling-element bearings. The report gives a typical natural frequency around 35 kHz and a useful upper measurement range of approximately 20 kHz, with a lower frequency limit around 1 Hz.

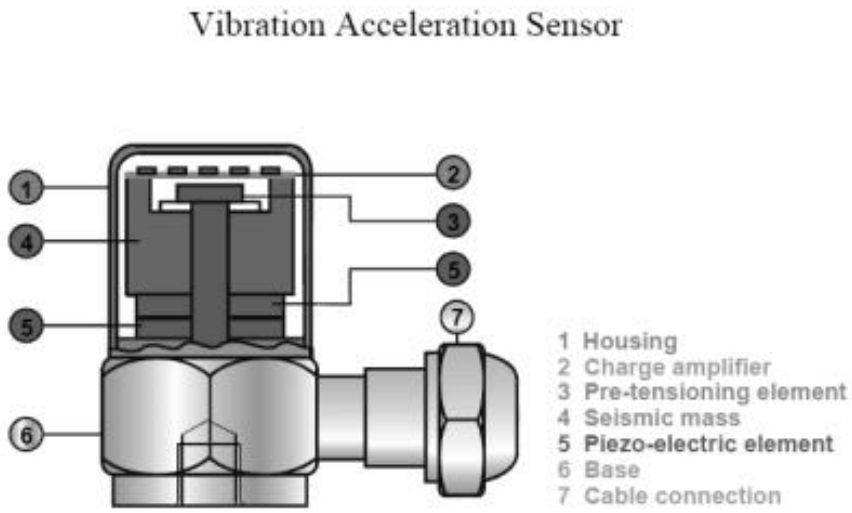


Figure 13. Internal construction of a piezoelectric accelerometer.

13. Actuators and Primary Airflow Control

The ECU uses sensor measurements, operating schedules, and limits to command actuators that improve performance and keep the engine within its safe envelope. The report identifies the VBV, VSV, HPTACC, and LPTACC systems as important actuators/control functions.

Variable Bleed Valve (VBV). Compressors can become unstable at low mass flow, potentially causing blade stall. Excessive pressure rise can lead to surge, which may cause severe engine disturbance. The VBV increases the amount of air bled from the compressor when required, helping prevent these conditions. Its position is commanded by the active ECU channel.

Variable Stator Vanes (VSV). Compressor blade rows include rotating rotor blades and stationary stator vanes. Changing the angle of the stator vanes adjusts the airflow through the compressor. The VSV actuator provides this adjustment.

Active Clearance Control (ACC). Engine heat and centrifugal loads cause components to expand and change the clearance between turbine-blade tips and the casing. Excessive clearance increases leakage and reduces efficiency. ACC varies the blade-tip clearance as operating conditions change. The report distinguishes low-pressure-turbine ACC (LPTACC) and high-pressure-turbine ACC (HPTACC).

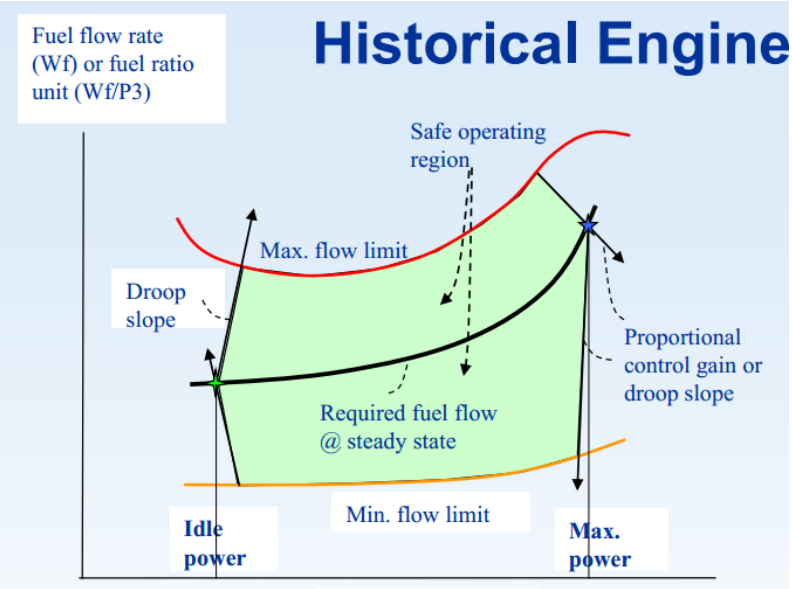


Figure 15. Fuel-flow operating limits and safe operating region.

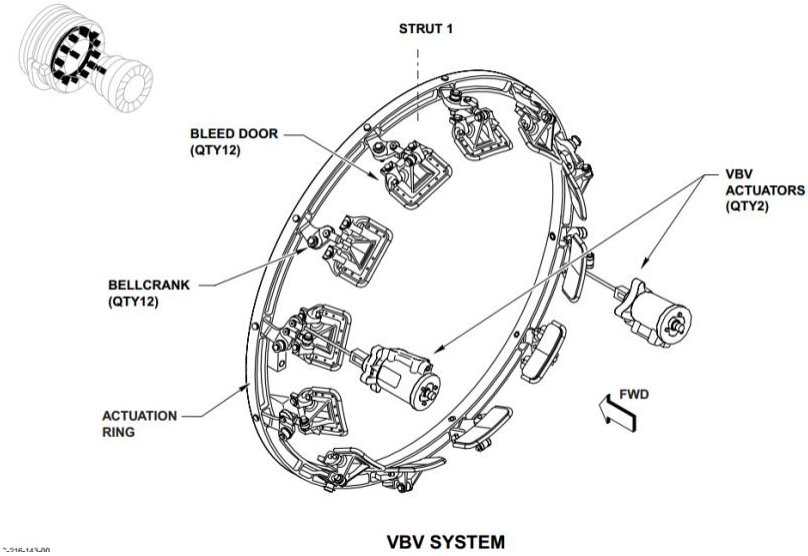
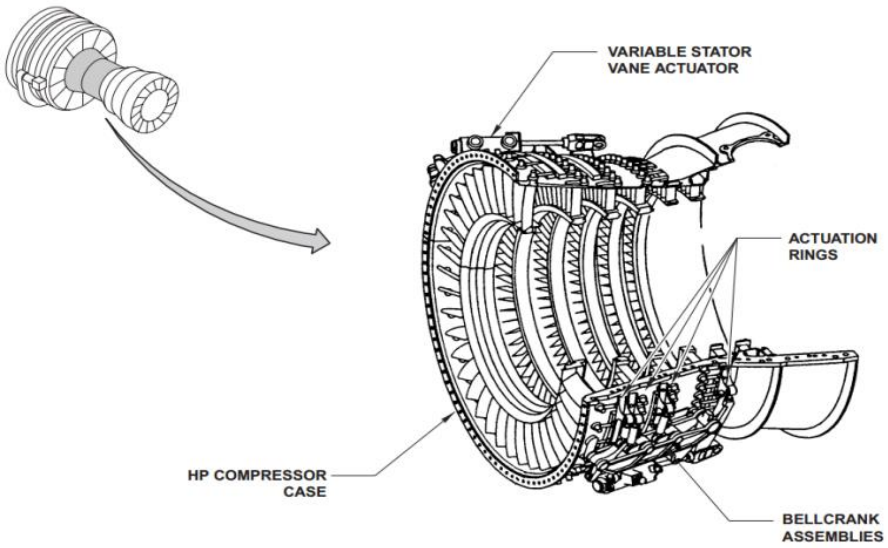
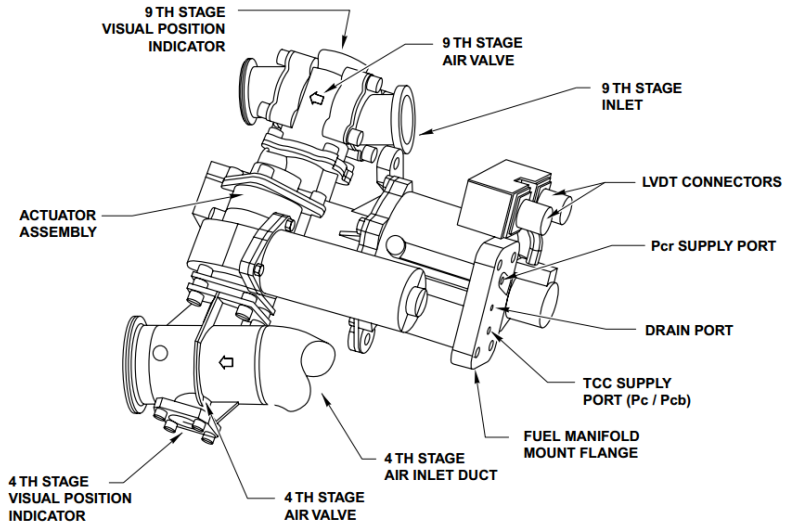


Figure 16. VBV system location and arrangement.



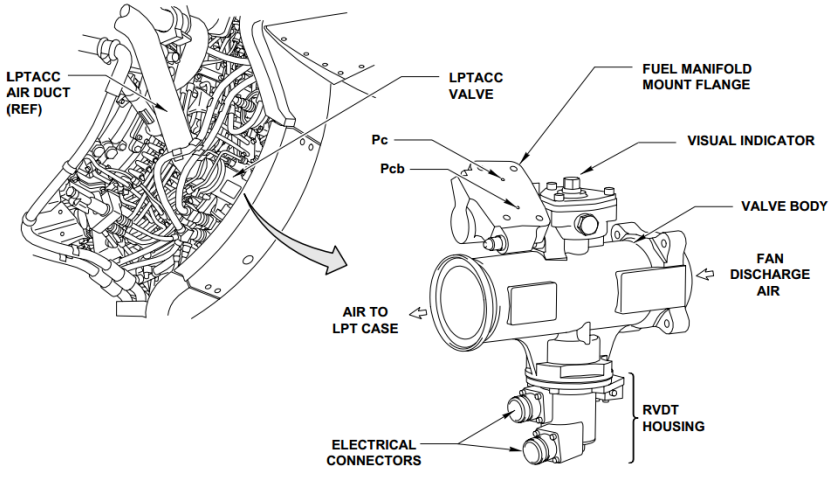
VSV SYSTEM LOCATION

Figure 17. VSV system location and actuator arrangement.



HPTACC VALVE

Figure 18. HPTACC valve assembly.



LPT ACTIVE CLEARANCE CONTROL VALVE

Figure 19. LPT active-clearance-control valve assembly.

14. Conclusion

The report surveys the principal electrical and control elements associated with the CFM56 engine. It describes the role of FADEC and the ECU in processing sensor measurements, coordinating engine actuators, managing thrust, and protecting the engine against operating-limit violations. It also reviews pressure and temperature sensing, shaft-speed measurement, vibration monitoring, and airflow/clearance actuators.

The main engineering theme is the interaction between sensing, control logic, and actuation in a safety-critical propulsion system. Redundant ECU channels, fault-priority logic, operating envelopes, and continuous monitoring support reliable operation while balancing performance, efficiency, and engine protection.

References

CFM (2006). CFM56-3 Regulation.

General Electric Aircraft Engines (revision 6, January 1998). Engine Systems (CF6-80C2BF): Customer Training Services.

Guo, J. C.-H. (2010). Control Design for a Generic Commercial Aircraft Engine. NASA.

Richards, D. R. (1984). The Infinite Line Pressure Probe. NASA.

Rohit K. Belapurkar, M. (2012). Stability and Performance of Propulsion Control Systems with Distributed Control Architectures and Failures. The Ohio State University.

Wikipedia (accessed August 6, 2018). FADEC. https://en.wikipedia.org/wiki/FADEC

Ghaffari, D. Measurement Instruments. K. N. Toosi University of Technology, Faculty of Mechanical Engineering.

MAPNA / TOGA (1397 / 2018). Identification of Sensors Used in the CFM56-5C2/F Engine. Engineering and Development, Aircraft Engine Maintenance section.

MAPNA / TOGA (1996). Engine Vibration Measurement System. Technical report, Engine Engineering and Maintenance section.

Translation note: Technical labels and reported values have been retained from the Persian source. A few source passages contain inconsistent sensor labels or wording; these are flagged where necessary rather than silently corrected.